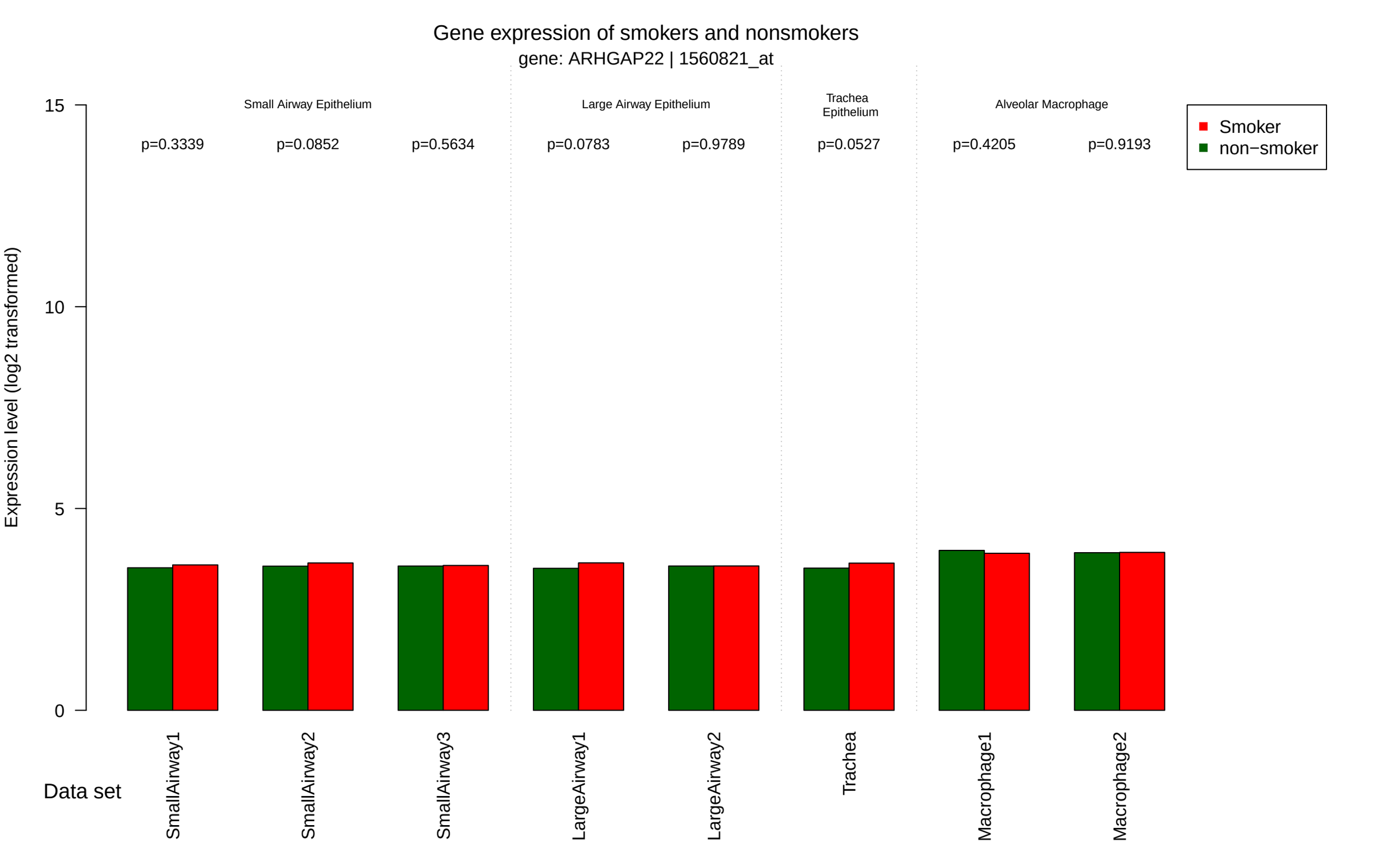
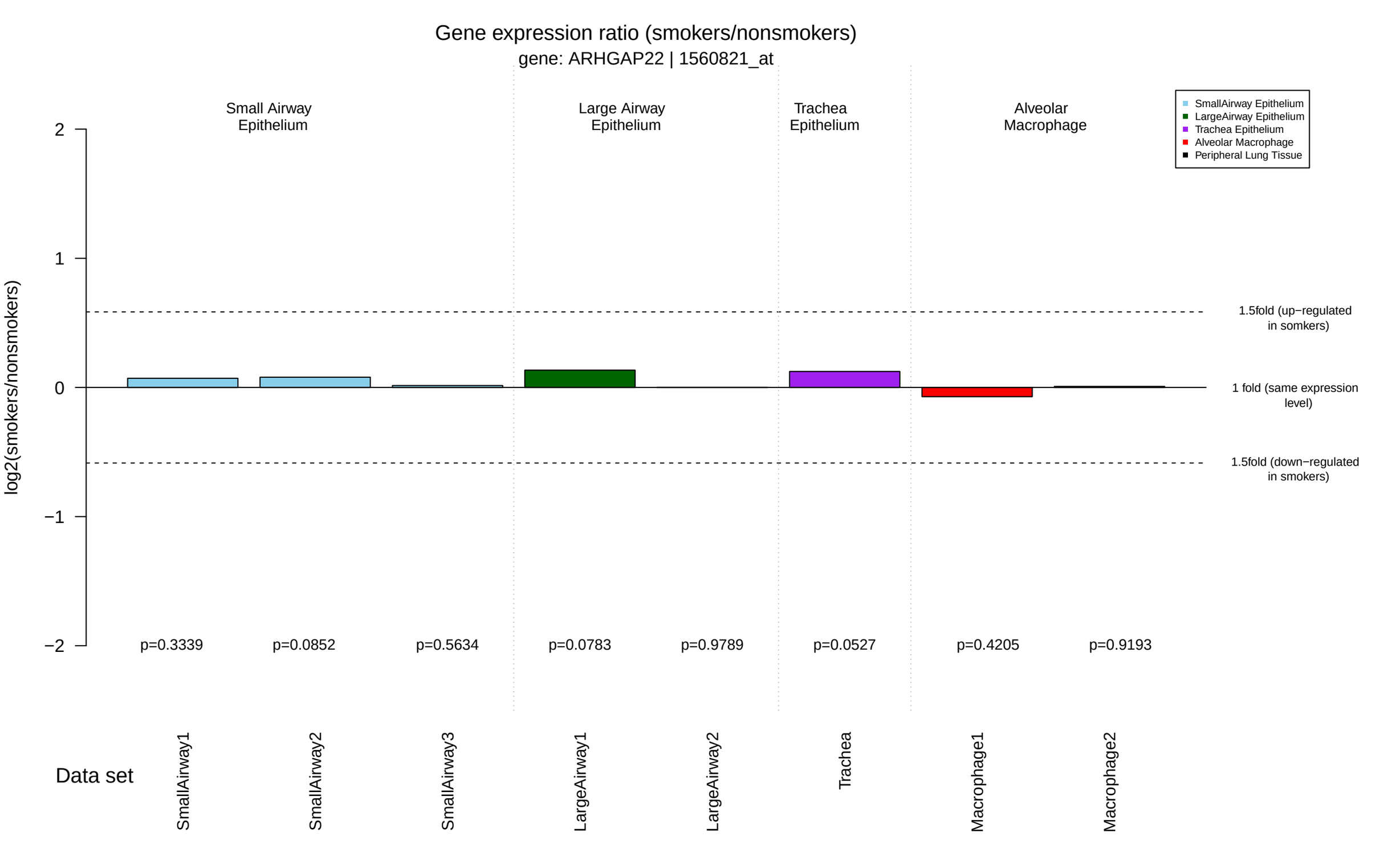
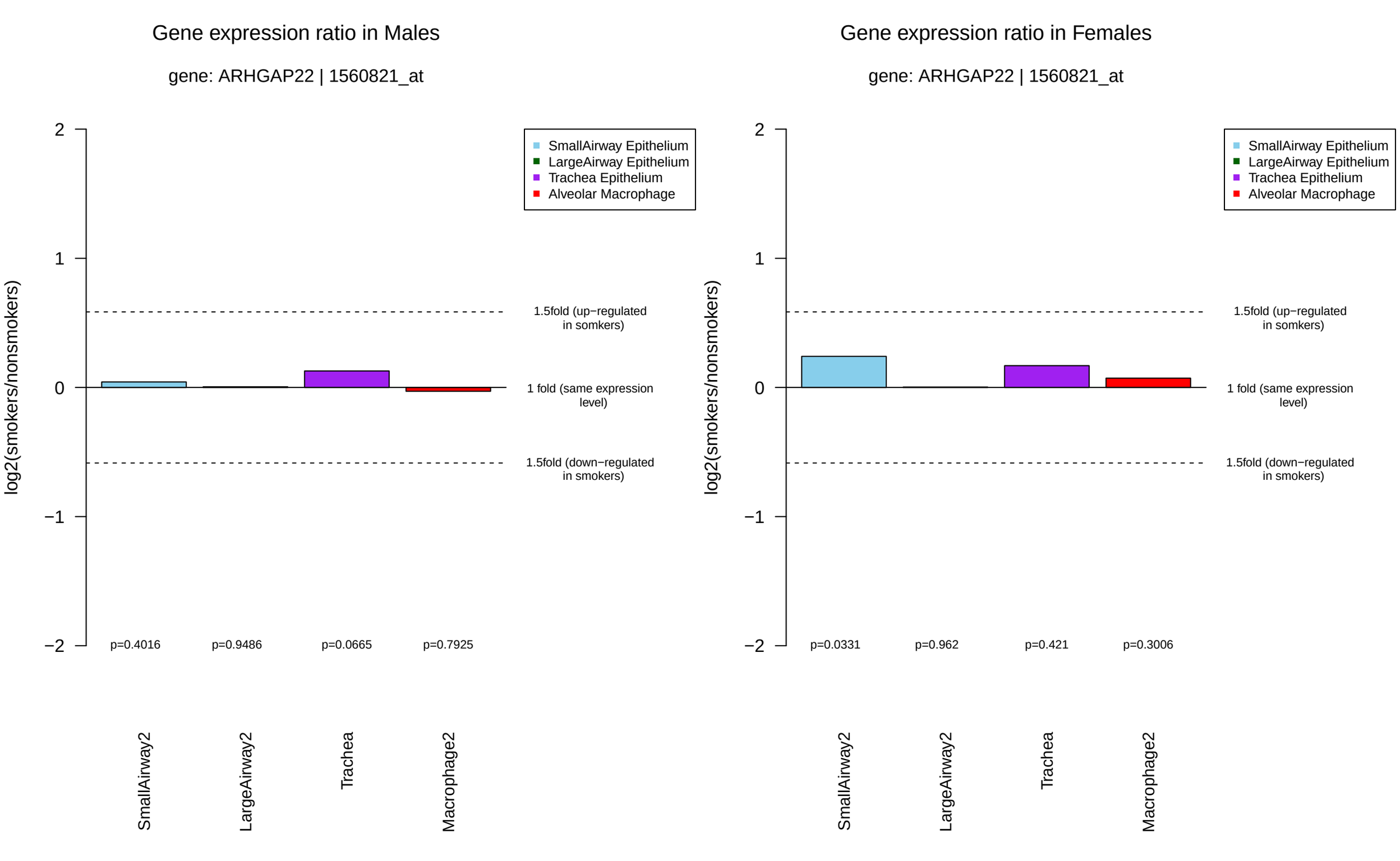
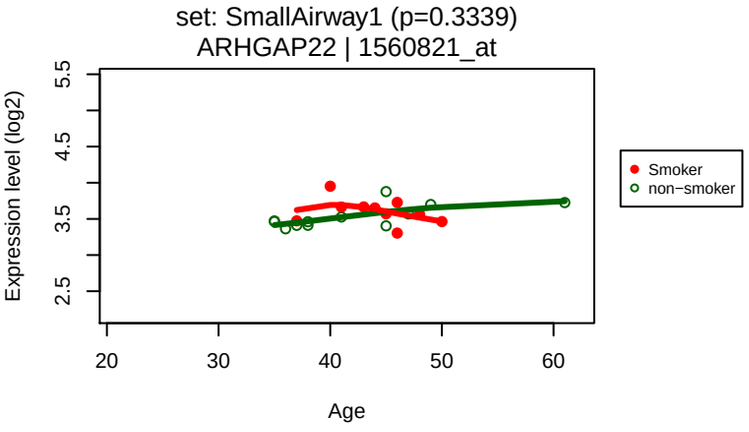
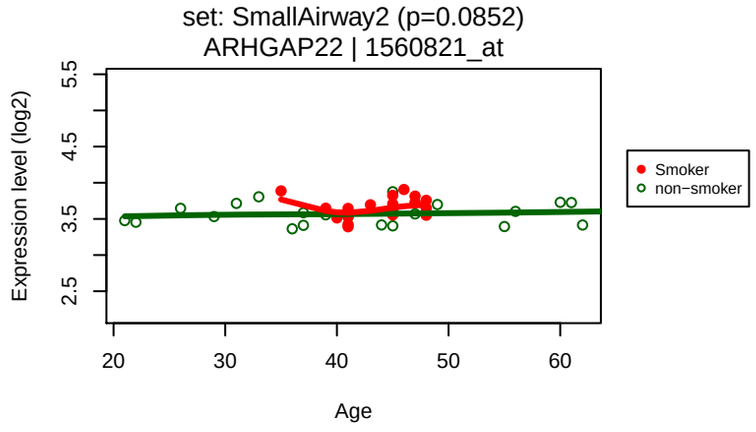
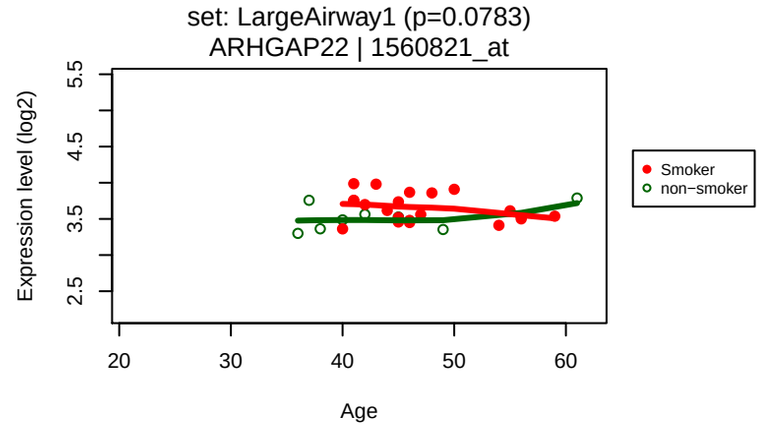

Gene expression of smokers and nonsmokers
gene: ARHGAP22 | 1560821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3339
p=0.0852
p=0.5634
p=0.0783
p=0.9789
p=0.0527
p=0.4205
p=0.9193
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

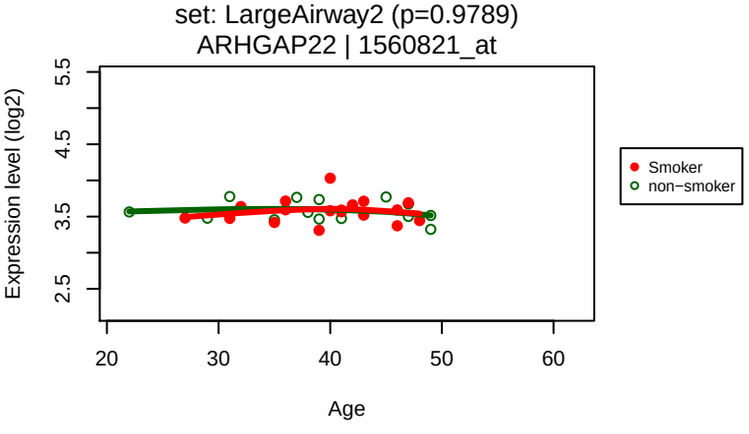

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP22 | 1560821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3339
p=0.0852
p=0.5634
p=0.0783
p=0.9789
p=0.0527
p=0.4205
p=0.9193
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

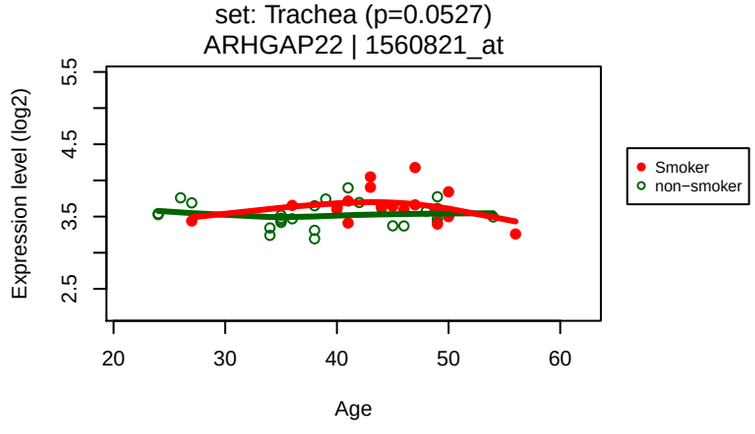

Gene expression ratio in Males
gene: ARHGAP22 | 1560821_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4016
p=0.9486
p=0.0665
p=0.7925
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP22 | 1560821_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0331
p=0.962
p=0.421
p=0.3006
SmallAirway2
LargeAirway2
Trachea
Macrophage2

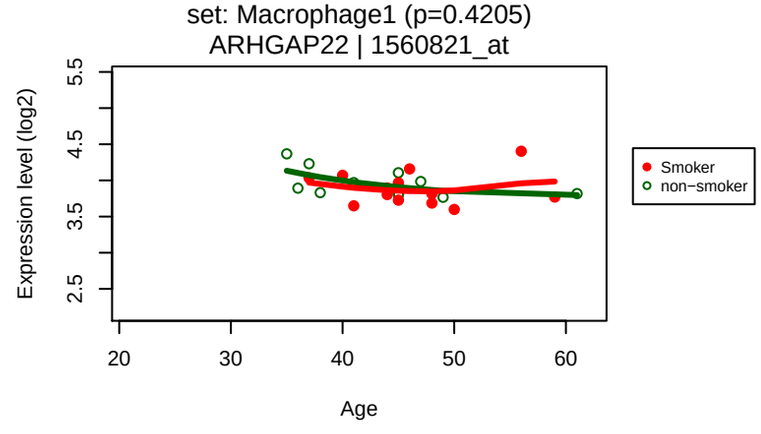

set: SmallAirway1 (p=0.3339)
ARHGAP22 | 1560821_at
Expression level (log2)
Age
Smoker
non−smoker

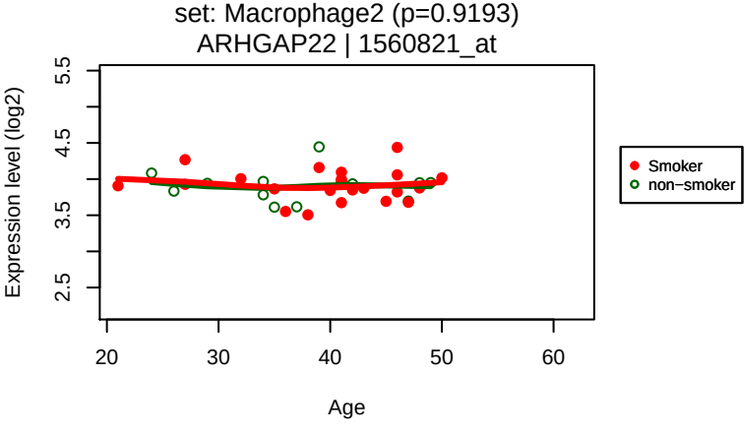

set: SmallAirway2 (p=0.0852)
ARHGAP22 | 1560821_at
Expression level (log2)
Age
Smoker
non−smoker

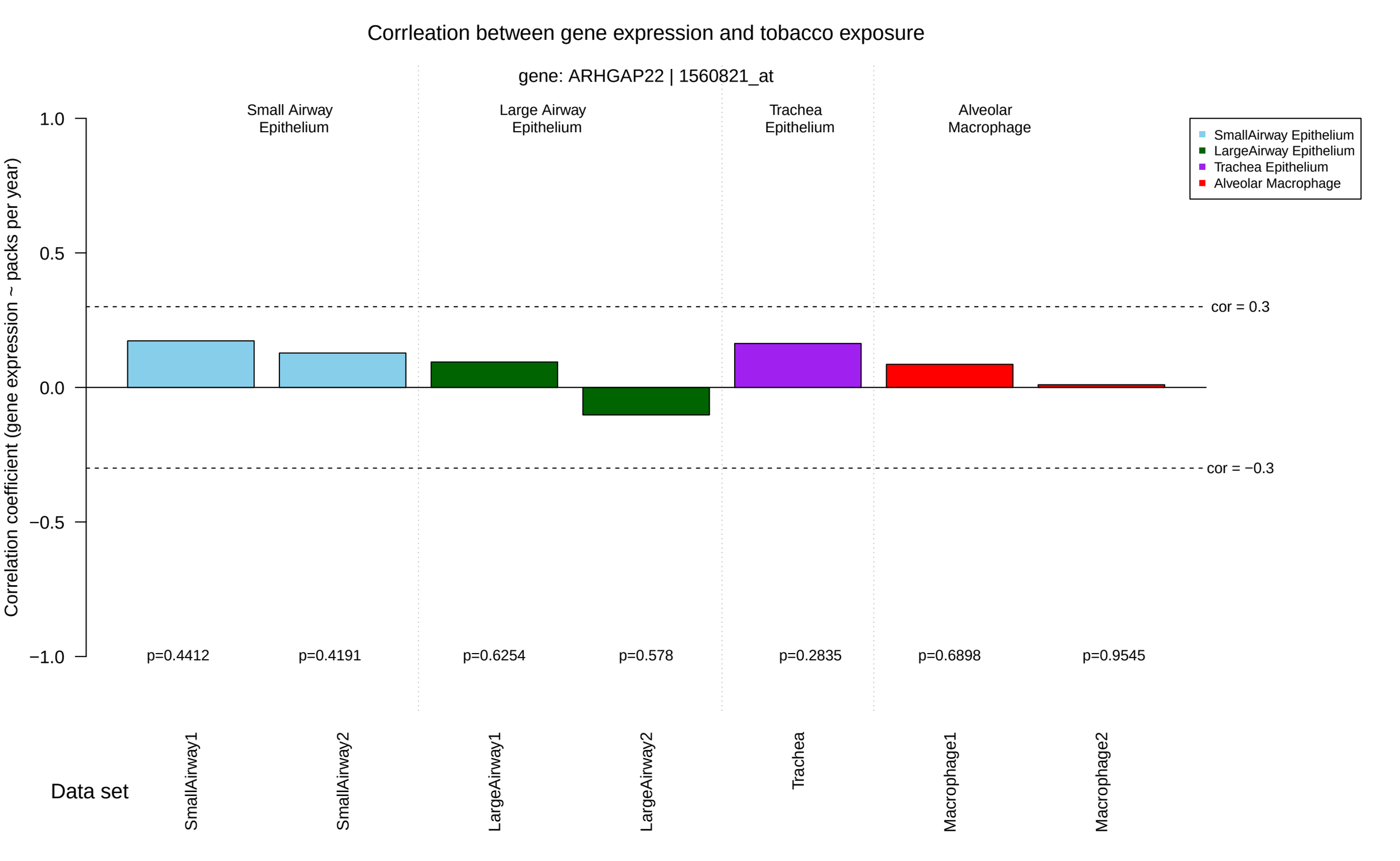

set: LargeAirway1 (p=0.0783)
ARHGAP22 | 1560821_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9789)
ARHGAP22 | 1560821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0527)
ARHGAP22 | 1560821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4205)
ARHGAP22 | 1560821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9193)
ARHGAP22 | 1560821_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP22 | 1560821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4412
p=0.4191
p=0.6254
p=0.578
p=0.2835
p=0.6898
p=0.9545
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2